

UNIT 2: 1D KINEMATICS • MASTERING QUADRATIC MOTION: $X_f = X_o + V_oT + \frac{1}{2}aT^2$

 THE 3 PHYSICAL COMPONENTS OF KINEMATIC DISPLACEMENT

$$X_f = X_o + V_o t + \frac{1}{2} a t^2$$

1. x_o (Origin Offset)
Starting coordinate position at $t = 0$ s. Standard reference value is 0 m unless scenario defines an initial mark.

2. $v_o \cdot t$ (Inertial Coasting)
Displacement covered if velocity remained constant (geometric rectangle on v-t graph).

3. $\frac{1}{2} \cdot a \cdot t^2$ (Acceleration Bonus)
Additional distance gained (or lost) due to changing velocity (geometric triangle on v-t).

 **Why is Time Squared (t^2)?**
Velocity increases with time ($v = at$), AND distance = avg speed $\times$ time $\rightarrow t \times t = t^2$!

PART 1: ACCELERATING FROM REST ($V_o = 0, X_o = 0 \Rightarrow X_f = \frac{1}{2}aT^2$)

TIER 1: FOUNDATION

Problem 1 • Top Fuel Dragster: A nitromethane dragster launches from a dead stop ($v_o = 0$ m/s) with a brutal, constant forward acceleration of $a = +28.0$ m/s² for $t = 3.20$ seconds.

GIVENS & UNKNOWN	FORMULA SELECTED	SUBSTITUTION WITH UNITS	CALCULATED OUTPUT
$v_o =$ $a =$ $t =$ $x_f = ?$	$x_f = x_o + v_o t + \frac{1}{2} a t^2$ (Simplify for $v_o = 0$)	$x_f = \frac{1}{2} (\quad) (\quad)^2$ $x_f =$	DISTANCE: m

Follow-up: The official NHRA dragstrip length is **305 meters** (1,000 ft). Has the car crossed the finish line at 3.20 s? [] YES [] NO — Explain:

Problem 2 • Superconducting Maglev Departure: A high-speed magnetic levitation bullet train accelerates smoothly from rest out of Tokyo Station at $a = +1.25$ m/s² for $t = 24.0$ seconds before reaching cruising speed. Calculate the distance traveled during this acceleration run.

GIVENS & UNKNOWN	FORMULA SETUP	SUBSTITUTION & ARITHMETIC STEPS	FINAL BOXED ANSWER
$v_o =$ $a =$ $t =$ $x_f = ?$	$x_f = \frac{1}{2} a t^2$	$x_f = \frac{1}{2} (\quad) (\quad)^2$ $x_f =$	DISTANCE: m

PART 2: ACCELERATING WITH INITIAL FORWARD SPEED ($V_o > 0, A > 0$)

TIER 2: INTERMEDIATE

Dual Component Strategy: When an object is already in motion, compute the two terms independently: *Coasting Distance* ($v_o \cdot t$) + *Acceleration Bonus* ($\frac{1}{2}at^2$), then sum them.

Problem 3 • Highway Passing Maneuver: A sports car cruises behind a semi-truck at a steady $v_o = +18.0$ m/s (40 mph). Opening the throttle, the driver accelerates at $a = +3.50$ m/s² for $t = 4.00$ seconds to pass.

GIVENS	COASTING TERM (V_oT)	ACCEL BONUS TERM ($\frac{1}{2}aT^2$)	TOTAL DISTANCE X_f
$v_o = +18.0$ m/s $a = +3.50$ m/s ² $t = 4.00$ s $x_f = ?$	$d_{\text{coast}} = (18.0 \text{ m/s})(4.00 \text{ s})$ $d_{\text{coast}} = \text{_____ m}$	$d_{\text{accel}} = \frac{1}{2} (3.50 \text{ m/s}^2)(4.00 \text{ s})^2$ $d_{\text{accel}} = \frac{1}{2} (3.50)(16.0)$ $d_{\text{accel}} = \text{_____ m}$	TOTAL X_f: m

Problem 4 • Commercial Jetliner Takeoff Roll: A twin-engine jet enters the runway threshold already rolling at $v_o = +10.0$ m/s (22 mph). The pilot sets full takeoff thrust, producing a constant acceleration of $a = +2.20$ m/s² for $t = 28.0$ seconds until rotation (liftoff). Calculate the total runway length consumed.

GIVENS & UNKNOWN	ALGEBRAIC FORMULA	STEP-BY-STEP SUBSTITUTION	LIFTOFF DISTANCE
$v_o =$ $a =$ $t =$ $x_f = ?$	$x_f = v_o t + \frac{1}{2} a t^2$	$x_f = (\quad) (\quad) + \frac{1}{2} (\quad) (\quad)^2$ $x_f = (\quad \text{m}) + (\quad \text{m})$	RUNWAY: m

PART 3: DECELERATION & EMERGENCY BRAKING (A < 0)

TIER 3: SIGN DISCIPLINE

⚠ Crucial Sign Rule: When brakes apply, acceleration points backwards opposite to motion (**a < 0**). The $\frac{1}{2}at^2$ term becomes *negative* and **subtracts** distance from what the vehicle would have coasted!

Problem 5 • Interstate Emergency Braking: A sedan travels at highway speed $v_o = +32.0$ m/s (72 mph). The driver observes stopped traffic and slams the brakes, causing an emergency deceleration of $a = -6.50$ m/s² for $t = 4.00$ seconds.

GIVENS (WATCH SIGNS!)	EQUATION SETUP	CALCULATION STEPS	SKID DISTANCE x_f
$v_o = +32.0$ m/s $a = -6.50$ m/s ² $t = 4.00$ s $x_f = ?$	$x_f = v_o t + \frac{1}{2}at^2$ (Note minus sign!)	$x_f = (32.0)(4.00) + \frac{1}{2}(-6.50)(4.00)^2$ $x_f = (128.0 \text{ m}) - (52.0 \text{ m})$	BRAKING x_f: m

Problem 6 • Aircraft Carrier Arrested Trap: An F/A-18 Hornet touches down on the flight deck at $v_o = +70.0$ m/s (157 mph). The arresting tailhook catches the cross-deck pendant cable wire, exerting violent deceleration of $a = -24.5$ m/s² for $t = 2.40$ seconds. Calculate the flight deck trap distance consumed.

GIVENS & UNKNOWN	EQUATION SETUP	STEP-BY-STEP SUBSTITUTION	TRAP DISTANCE
$v_o =$ $a =$ $t =$ $x_f = ?$	$x_f = v_o t + \frac{1}{2}at^2$	$x_f = (\quad)(\quad) + \frac{1}{2}(\quad)$ $(\quad)^2$ $x_f = (\quad \text{m}) - (\quad \text{m})$	DECK RUN: m

PART 4: MULTI-STAGE KINEMATICS • DRIVER REACTION & COLLISION AVOIDANCE

TIER 4: SYNTHESIS

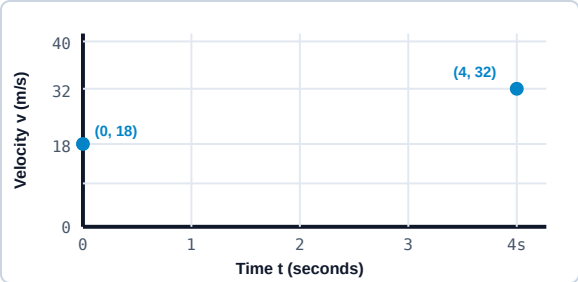
Problem 7 • Wildlife Obstacle Avoidance: An autonomous vehicle cruises along a dark mountain road at $v_o = +24.0$ m/s (54 mph). At $t = 0$ s, sensors detect a fallen tree trunk blocking the road 62.0 meters ahead!

STAGE 1: NEURAL / SENSOR DELAY	STAGE 2: EMERGENCY ABS BRAKING	STAGE 3: DECISION & MARGIN
The computer processes the lidar image for $t_1 = 0.50$ s. During this delay, $a = 0$ m/s ² (constant speed). $d_1 = v_o \cdot t_1 = (24.0 \text{ m/s})(0.50 \text{ s})$ $d_1 = \text{_____ m}$	Brakes engage at maximum friction: $a = -8.00$ m/s ² for $t_2 = 3.00$ s until coming to a complete rest. $d_2 = v_o t_2 + \frac{1}{2}at_2^2$ $d_2 = (24.0)(3.00) + \frac{1}{2}(-8.00)(3.00)^2$ $d_2 = \text{_____ m}$	Total stopping distance: $d_{\text{total}} = d_1 + d_2 = \text{_____ m}$ Does the car stop before the tree (62.0 m)? [] SAFELY STOPS [] CRASHES Safety Clearance Margin: $\pm \text{_____ m}$

PART 5: GEOMETRIC INTEGRATION • CONNECTING $x_f = v_o T + \frac{1}{2}AT^2$ TO VELOCITY-TIME GRAPHS

DOK 3

Problem 8 • Geometric Area Proof: Refer back to Problem 3 (Sports car: $v_o = 18$ m/s, accelerating at $a = 3.5$ m/s² for 4.0 s to $v_f = 32$ m/s). On the grid below, draw the line from (0 s, 18 m/s) to (4 s, 32 m/s). Shade the Rectangle and Triangle underneath.



1. Bottom Shaded Rectangle Area (Inertial Coasting): $\text{Area}_{\text{rect}} = \text{base} \times \text{height} = (4.0 \text{ s}) \times (18.0 \text{ m/s}) = \text{_____ meters}$
2. Top Shaded Triangle Area (Acceleration Bonus): $\text{Area}_{\text{tri}} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times (4.0 \text{ s}) \times (32.0 - 18.0 \text{ m/s}) = \frac{1}{2}(4)(14) = \text{_____ meters}$
✓ Synthesis: Total Area = $\text{Area}_{\text{rect}} + \text{Area}_{\text{tri}} = \text{_____ m} \equiv \text{Exact Algebraic Match!}$